

Introduction

Two-dimensional type-II nonuniform discrete Fourier transform (NUDFT):

$$f_j = \sum_{k^{[x]}=0}^{n^{[x]}-1} \sum_{k^{[y]}=0}^{n^{[y]}-1} c_{k^{[x]},k^{[y]}} e^{-2\pi i(k^{[x]}x_j + k^{[y]}y_j)}, \quad 0 \leq j \leq M-1,$$

- Sample points: $(x_j, y_j) \in [0, 1]^2$;
- Frequencies: $(k^{[x]}, k^{[y]}) \in \{0, 1, \dots, n^{[x]}-1\} \times \{0, 1, \dots, n^{[y]}-1\}$;
- Coefficients: $c_{k^{[x]},k^{[y]}} \in \mathbb{C}$;
- Target values: $f_j \in \mathbb{C}$;
- Number of sample points M , number of frequencies $N = n^{[x]}n^{[y]}$.

NUDFT matrix:

$$\mathbf{A} = \left(e^{-2\pi i(k^{[x]}x_j + k^{[y]}y_j)} \right)_{j, (k^{[x]}, k^{[y]})} \in \mathbb{C}^{M \times N}.$$

Problem:

$$\min_{\mathbf{c} \in \mathbb{C}^N} \|\mathbf{A}\mathbf{c} - \mathbf{f}\|$$

A Review of the Method in 1D Case [1]

1D NUDFT matrix:

$$\mathbf{A} = \left(e^{-2\pi i k x_j} \right)_{j,k} \in \mathbb{C}^{M \times N} \quad \text{with} \quad x_j \in [0, 1).$$

1D DFT matrix:

$$\mathbf{F} = \left(\mathbf{F}(k, \ell) = e^{-2\pi i k \ell / N} \right)_{k, \ell} \in \mathbb{C}^{N \times N}.$$

The matrix $\mathbf{G} = \mathbf{A}\mathbf{F}^{-1}$ can be expressed as a kernel matrix:

$$\mathbf{G}(j, k) = \frac{1}{N} \frac{(\gamma_j / \xi_k)^N - 1}{(\gamma_j / \xi_k) - 1} =: \Psi(\gamma_j, \xi_k).$$

- $\gamma_j = e^{-2\pi i x_j} \in \mathbb{S}$;
- $\xi_k = e^{-2\pi i k / N} \in \mathbb{S}$.

↪ Approximate $\mathbf{G}$ by a hierarchically semiseparable (HSS) matrix.

↪ Use the HSS least-squares solver (URV factorization).

Theorem (Numerical rank of HSS blocks). Let $\mathcal{J} = \{0, 1, \dots, M-1\}$ and $\mathcal{K} = \{0, 1, \dots, N-1\}$. Suppose that $[a, b] \subset [0, 1)$ is an interval and $\mathcal{R} = \{j : x_j \in [a, b]\} \subset \mathcal{J}$ and $\mathcal{C} = \{k : k/N \in [a, b]\} \subset \mathcal{K}$ are the row and column index sets corresponding to $[a, b]$. Then for $0 < \varepsilon < 1$, the HSS row or column block of $\mathbf{G}$ with respect to $\mathcal{R}$ or $\mathcal{C}$ has a ε -rank at most $\mathcal{O}(\log(1/\varepsilon) \log n_c)$ where $n_c = |\mathcal{C}|$.

Low-Rank Property of the NUDFT Matrix

NUDFT matrices in x and y direction:

$$\mathbf{A}^{[x]} = \left(e^{-2\pi i k^{[x]} x_j} \right)_{j, k^{[x]}} \in \mathbb{C}^{M \times n^{[x]}} \quad \text{and} \quad \mathbf{A}^{[y]} = \left(e^{-2\pi i k^{[y]} y_j} \right)_{j, k^{[y]}} \in \mathbb{C}^{M \times n^{[y]}}.$$

Then

$$\mathbf{A} = \mathbf{A}^{[x]} \bullet \mathbf{A}^{[y]} = \begin{bmatrix} \mathbf{A}^{[x]}(0, :) \otimes \mathbf{A}^{[y]}(0, :) \\ \mathbf{A}^{[x]}(1, :) \otimes \mathbf{A}^{[y]}(1, :) \\ \vdots \\ \mathbf{A}^{[x]}(M-1, :) \otimes \mathbf{A}^{[y]}(M-1, :) \end{bmatrix} \in \mathbb{C}^{M \times N}.$$

DFT matrices in x and y direction:

$$\mathbf{F}^{[x]} \in \mathbb{C}^{n^{[x]} \times n^{[x]}} \quad \text{and} \quad \mathbf{F}^{[y]} \in \mathbb{C}^{n^{[y]} \times n^{[y]}}.$$

Let $\mathbf{F} = \mathbf{F}^{[x]} \otimes \mathbf{F}^{[y]} \in \mathbb{C}^{N \times N}$ be the 2D DFT matrix, then

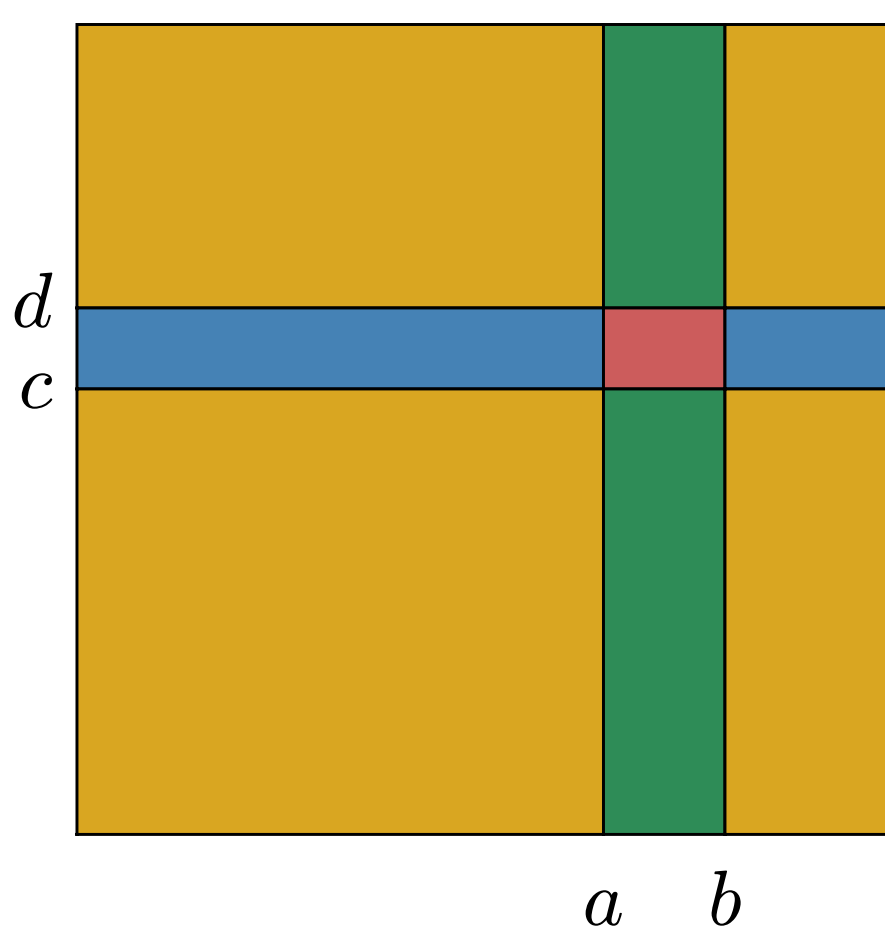
$$\mathbf{G} = \mathbf{A}\mathbf{F}^{-1} = (\mathbf{A}^{[x]} \mathbf{F}^{[x], -1}) \bullet (\mathbf{A}^{[y]} \mathbf{F}^{[y], -1}) = \mathbf{G}^{[x]} \bullet \mathbf{G}^{[y]},$$

whose entries can be expressed as

$$\mathbf{G}(j, (k^{[x]}, k^{[y]})) = \Psi^{[x]}(\gamma_j^{[x]}, \xi_{k^{[x]}}^{[x]}) \Psi^{[y]}(\gamma_j^{[y]}, \xi_{k^{[y]}}^{[y]}) = \Psi(\gamma_j, \xi_{(k^{[x]}, k^{[y]})}).$$

- $\gamma_j^{[x]} = e^{-2\pi i x_j}$, $\gamma_j^{[y]} = e^{-2\pi i y_j}$, $\gamma_j = (\gamma_j^{[x]}, \gamma_j^{[y]}) \in \mathbb{S}^2$;
- $\xi_{k^{[x]}}^{[x]} = e^{-2\pi i k^{[x]} / n^{[x]}}$, $\xi_{k^{[y]}}^{[y]} = e^{-2\pi i k^{[y]} / n^{[y]}}$, $\xi_{(k^{[x]}, k^{[y]})} = (\xi_{k^{[x]}}^{[x]}, \xi_{k^{[y]}}^{[y]}) \in \mathbb{S}^2$.

Lemma (Rank of the matrix face-splitting product). $\text{rank}(\mathbf{A}_1 \bullet \mathbf{A}_2) \leq \text{rank}(\mathbf{A}_1) \text{rank}(\mathbf{A}_2)$.



Submatrix of $\mathbf{G}$:

- $\mathcal{R}$: Row index set corresponding to row points in $[a, b] \times [c, d]$;
- $\mathcal{C} = \mathcal{C}^{[x]} \times \mathcal{C}^{[y]}$. Column index set corresponding to column points in $[a, b] \times [c, d]$;
- $\mathcal{C}^c = \mathcal{K} \setminus \mathcal{C} = \mathcal{C}_b \cup \mathcal{C}_g \cup \mathcal{C}_y$, corresponding to the blue, green and yellow boxes respectively.

The submatrix can be partitioned as:

$$\mathbf{G}(\mathcal{R}, \mathcal{C}^c) = \begin{bmatrix} \mathbf{G}(\mathcal{R}, \mathcal{C}_b) & \mathbf{G}(\mathcal{R}, \mathcal{C}_g) & \mathbf{G}(\mathcal{R}, \mathcal{C}_y) \end{bmatrix},$$

where

$$\begin{aligned} \mathbf{G}(\mathcal{R}, \mathcal{C}_b) &= \underbrace{\mathbf{G}^{[x]}(\mathcal{R}, \mathcal{C}^{[x], c})}_{\text{low-rank}} \bullet \underbrace{\mathbf{G}^{[y]}(\mathcal{R}, \mathcal{C}^{[y]})}_{\text{full-rank}}, \\ \mathbf{G}(\mathcal{R}, \mathcal{C}_g) &= \underbrace{\mathbf{G}^{[x]}(\mathcal{R}, \mathcal{C}^{[x]})}_{\text{full-rank}} \bullet \underbrace{\mathbf{G}^{[y]}(\mathcal{R}, \mathcal{C}^{[y], c})}_{\text{low-rank}}, \\ \mathbf{G}(\mathcal{R}, \mathcal{C}_y) &= \underbrace{\mathbf{G}^{[x]}(\mathcal{R}, \mathcal{C}^{[x], c})}_{\text{low-rank}} \bullet \underbrace{\mathbf{G}^{[y]}(\mathcal{R}, \mathcal{C}^{[y], c})}_{\text{low-rank}}. \end{aligned}$$

Therefore, intuitively,

$$\text{rank}_\varepsilon(\mathbf{G}(\mathcal{R}, \mathcal{C}^c)) \approx \mathcal{O}(\log(1/\varepsilon) n_c \log n_c), \quad n_c = \max\left\{\left|\mathcal{C}^{[x]}\right|, \left|\mathcal{C}^{[y]}\right|\right\}.$$

↪ $\mathbf{G}$ can be compressed in HSS format whose HSS rank scales as $\mathcal{O}(\sqrt{N} \log N)$.

Face-Splitting Product of HSS Trees and HSS Matrices

Definition (Face-splitting product of HSS trees). Let $\mathbf{T}^{[x]}$ and $\mathbf{T}^{[y]}$ be two HSS trees of row and column index sets $\mathcal{J}$ and $\mathcal{K}^{[x]}$ and $\mathcal{J}$ and $\mathcal{K}^{[y]}$ respectively. Suppose that $\mathbf{T}^{[x]}$ and $\mathbf{T}^{[y]}$ are both full trees of the same maximum level, then the face-splitting product of $\mathbf{T}^{[x]}$ and $\mathbf{T}^{[y]}$, denoted by $\mathbf{T} = \mathbf{T}^{[x]} \bullet \mathbf{T}^{[y]}$, is defined as follows.

- (1) The root of $\mathbf{T}$ is $\rho = (\rho^{[x]}, \rho^{[y]})$, where $\rho^{[x]}$ and $\rho^{[y]}$ are the roots of $\mathbf{T}^{[x]}$ and $\mathbf{T}^{[y]}$ respectively.
- (2) If $\tau^{[x]}$ is a leaf and $\tau^{[y]}$ is a nonleaf, then $\tau = (\tau^{[x]}, \tau^{[y]})$ is a nonleaf node of $\mathbf{T}$ and its children are given by $\text{ch}(\tau) = \{(\tau^{[x]}, \alpha^{[y]}) : \alpha^{[y]} \in \text{ch}(\tau^{[y]})\}$.
- (3) If $\tau^{[x]}$ and $\tau^{[y]}$ are both leaves, then $\tau = (\tau^{[x]}, \tau^{[y]})$ is a leaf node of $\mathbf{T}$.
- (4) Each node $\tau = (\tau^{[x]}, \tau^{[y]})$ is associated with the row index set $\mathcal{J}_\tau = \mathcal{J}_{\tau^{[x]}} \cap \mathcal{J}_{\tau^{[y]}}$ and the column index set $\mathcal{K}_\tau = \mathcal{K}_{\tau^{[x]}} \times \mathcal{K}_{\tau^{[y]}}$.

Theorem (Face-splitting product of HSS trees is an HSS tree). $\mathbf{T} = \mathbf{T}^{[x]} \bullet \mathbf{T}^{[y]}$ is an HSS tree.

Theorem (Face-splitting product of HSS matrices is an HSS matrix). Let $\mathbf{H}^{[x]}$ and $\mathbf{H}^{[y]}$ be two HSS matrices of $\mathbf{T}^{[x]}$ and $\mathbf{T}^{[y]}$, then $\mathbf{H} = \mathbf{H}^{[x]} \bullet \mathbf{H}^{[y]}$ is an HSS matrix of $\mathbf{T} = \mathbf{T}^{[x]} \bullet \mathbf{T}^{[y]}$.

The Construction of the HSS Approximation

Lemma (Interpolative decomposition). Let $\mathbf{A}$ be an $m \times n$ matrix with rank $r \leq \min\{m, n\}$. Then there exist a subset $\widehat{\mathcal{J}} \subset \{1, 2, \dots, m\}$ of size r and a matrix $\mathbf{L} \in \mathbb{C}^{m \times r}$ such that $\mathbf{A} = \mathbf{L}\mathbf{A}(\widehat{\mathcal{J}}, :)$.

For each node τ , let $\mathcal{P}_\tau$ be the set of proxy points such that

$$\mathbf{G}(\mathcal{J}_\tau, \mathcal{K}_\tau^c) = \Psi(\mathcal{L}_{\mathcal{J}_\tau}, \mathcal{W}_{\mathcal{K}_\tau^c}) \approx \Psi(\mathcal{L}_{\mathcal{J}_\tau}, \mathcal{P}_\tau) \mathbf{Q}_\tau^*,$$

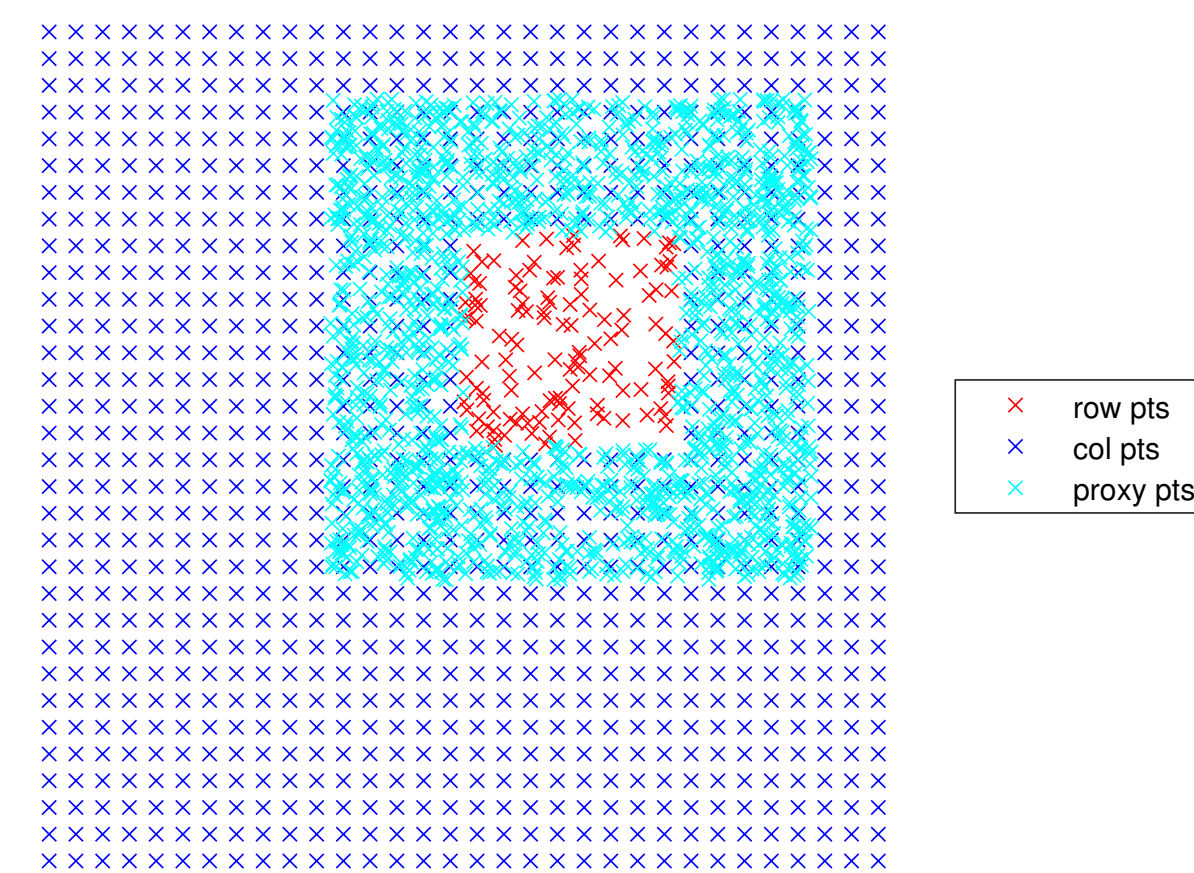
for some matrix $\mathbf{Q}_\tau$. Then, if we have the row ID

$$\Psi(\mathcal{L}_{\mathcal{J}_\tau}, \mathcal{P}_\tau) \approx \mathbf{L}_\tau \Psi(\mathcal{L}_{\mathcal{J}_\tau}, \mathcal{P}_\tau),$$

it follows that

$$\mathbf{G}(\mathcal{J}_\tau, \mathcal{K}_\tau^c) \approx \Psi(\mathcal{L}_{\mathcal{J}_\tau}, \mathcal{W}_{\mathcal{K}_\tau^c}) \approx \mathbf{L}_\tau \Psi(\mathcal{L}_{\mathcal{J}_\tau}, \mathcal{P}_\tau) \mathbf{Q}_\tau^*,$$

which gives the row ID of $\mathbf{G}(\mathcal{J}_\tau, \mathcal{K}_\tau^c)$.



Approximate Direct Inversion of 2D Type-II NUDFT

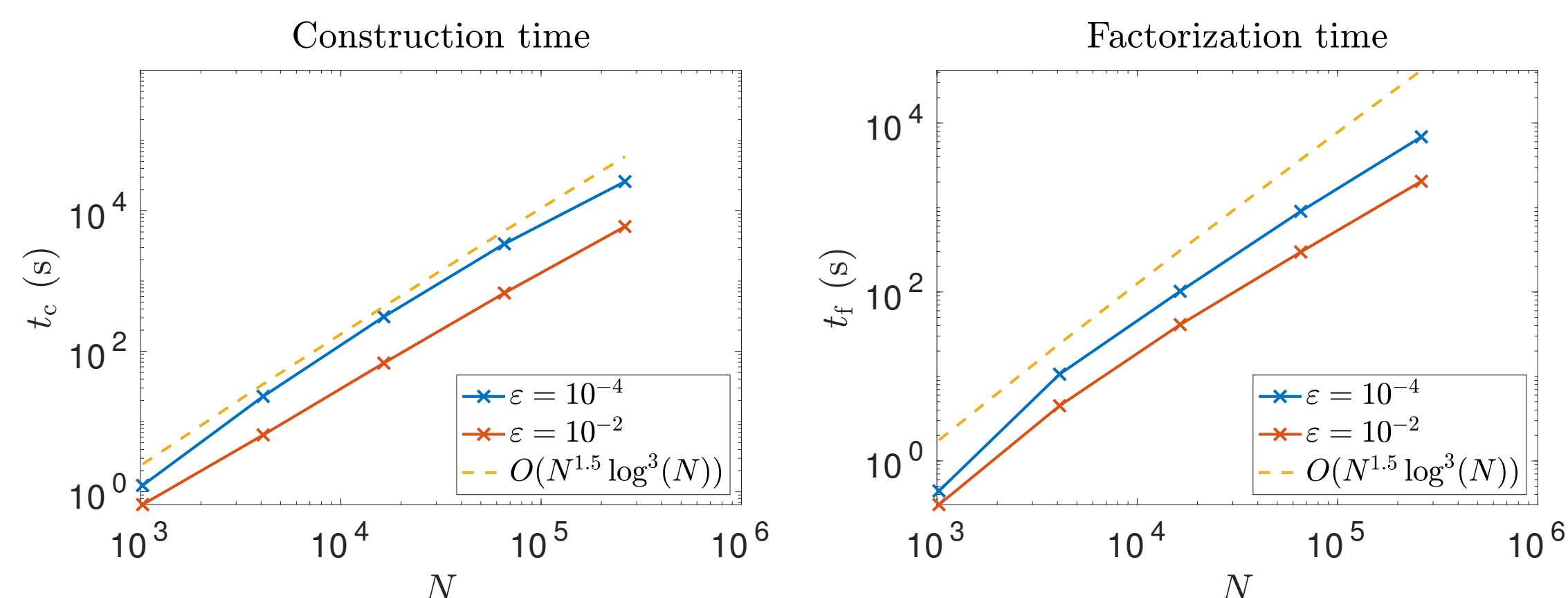
Algorithm A direct solver for type-II 2D NUDFT problem

- 1: Construct the HSS approximation $\widehat{\mathbf{G}}$ of $\mathbf{G} = \mathbf{A}\mathbf{F}^{-1}$ using the kernel matrix expression.
- 2: Compute the URV factorization of $\widehat{\mathbf{G}}$.
- 3: Compute $\mathbf{c} = \mathbf{F}^{-1} \widehat{\mathbf{G}}^\dagger \mathbf{f}$ by the URV solution and two-dimensional iFFT.

Complexities:

- **Offline stage** (construction and factorization): $\mathcal{O}(M + N^{3/2} \log^3 N)$;
- **Online stage** (solution): $\mathcal{O}(M + N \log^3 N)$.

Numerical Results



N	M	method	t_{pre}	t_{iter}	n_{iter}	r_s	e_s
32^2	2397	lsqr	-	1.3e+00	470	8.9e-13	6.8e-12
		plsqr ($\varepsilon = 10^{-2}$)	9.6e-01	1.9e-01	9	2.9e-13	9.9e-12
		plsqr ($\varepsilon = 10^{-4}$)	1.7e+00	1.5e-01	4	7.7e-16	6.5e-15
64^2	11620	lsqr	-	2.9e+00	500	3.8e-03	4.0e-03
		plsqr ($\varepsilon = 10^{-2}$)	1.1e+01	2.2e+00	21	1.5e-11	1.6e-11
		plsqr ($\varepsilon = 10^{-4}$)	3.3e+01	1.4e+00	5	8.9e-11	9.3e-11
128^2	54373	lsqr	-	5.6e+00	500	2.2e-05	1.6e-03
		plsqr ($\varepsilon = 10^{-2}$)	1.1e+02	2.8e+01	38	2.6e-13	1.4e-11
		plsqr ($\varepsilon = 10^{-4}$)	4.1e+02	1.5e+01	10	2.2e-13	6.7e-11
256^2	249074	lsqr	-	1.7e+01	500	1.9e-03	1.8e-03
		plsqr ($\varepsilon = 10^{-2}$)	9.7e+02	1.3e+02	37	1.3e-11	1.2e-11
		plsqr ($\varepsilon = 10^{-4}$)	4.3e+03	1.0e+02	12	5.7e-11	5.5e-11
512^2	1122019	lsqr	-	6.5e+01	500	4.8e-04	4.6e-04
		plsqr ($\varepsilon = 10^{-2}$)	8.0e+03	4.2e+02	24	2.2e-12	2.1e-12
		plsqr ($\varepsilon = 10^{-4}$)	3.3e+04	4.2e+02	10	2.5e-12	2.4e-12

References

- [1] Heather Wilber, Ethan N. Epperly, and Alex H. Barnett.
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My Homepage

<https://jingyuliumath.github.io>